

COMMON GROUND | Investment memo

Committee review draft | 5 October 2026 | France / Paris-Saclay study region

Decision: send back for more evidence

Do not authorize the full 25 MW construction commitment. Fund only a separately budgeted demand and site study. Seek comparable lease offers and a smaller first-phase option. No construction budget, supplier commitment or final build/lease/hybrid choice is established.

What is being reviewed

The original proposal is 20 MW IT at PUE 1.25: 25 MW facility power and 219 GWh/year at continuous full load. Members have expressed interest but have not signed long-term compute commitments. Grid connection price, upgrades and energization date remain unknown. The Paris-Saclay region is a provisional study location, not a selected parcel.

Planning demand is 28 million productive GPU-hours/year, with 5,000 concurrent equivalent GPUs and a 70% effective scheduling ceiling. At 2 kW allocated IT per equivalent GPU, the calculated requirement is 10 MW IT / 12.5 MW facility capacity. Hardware equivalence, timing, network performance and member demand remain unverified. The smaller result is a revision to investigate, not proof of actual need.

Three alternatives; one assumed workload

Full build owns the original 20 MW IT fleet. Hybrid owns 8 MW IT and contracts for the residual workload. Lease buys equivalent compute with a minimum commitment. All three serve the same assumed productive demand; national electricity price is a reference, not a site offer.

Base case	Before opening	Year 1 OPEX	EUR/GPU-h	Capital at risk
Build	EUR 480.0m	EUR 33.0m	4.45	EUR 402.0m
Lease	EUR 0.0m	EUR 105.1m	3.75	EUR 105.1m
Hybrid	EUR 192.0m	EUR 76.3m	4.03	EUR 223.9m

Screening model: ten years, constant euros, no discounting. Facility/grid capital and GPU purchases/replacement are separate. Cost per productive GPU-hour includes net capital, operating and financing outflows less terminal recovery. France reference electricity: EUR 0.0614/kWh, Eurostat 2025-S2. Other costs are illustrative assumptions, not bids.

Stress cases and funding exposure

Full build	Before opening	Year 1 OPEX	EUR/GPU-h	Capital at risk
Base	EUR 480.0m	EUR 33.0m	4.45	EUR 402.0m
Grid +1 year	EUR 584.8m	EUR 94.0m	4.61	EUR 486.0m
Half use	EUR 480.0m	EUR 31.7m	8.80	EUR 402.0m

Delay assumes owned operation begins in year 2, with temporary leasing during year 1. Before-opening cash includes year 0 plus year 1 net outflows, including deferred GPUs and financing. Half use retains the fleet and fixed costs; idle IT still draws power. Capital at risk is unrecovered initial capital plus minimum/bridge lease exposure, not a probability-weighted loss. The website shows all nine combinations and annual cash flows.

Three findings most likely to change the decision

1. Committed demand. Obtain member GPU-hour and payment commitments, peak-cluster benchmarks, class calendars, availability/security needs and exit obligations. Adequate commitments at sustainable prices may support ownership; weak demand favors a smaller phase or leasing.

2. Deliverable power and site. Obtain a grid offer with date, upgrade cost, permits and cost allocation to other customers. Validate critical-load failure paths, 48-hour fuel logistics, cooling and water limits. A long delay or infeasible site may remove the build option.

3. Comparable supply bids. Compare construction, GPU procurement/finance and equivalent lease offers under the same workload, security, availability and replacement assumptions. Include unused-capacity and member-withdrawal risk. Lower modeled cost alone is insufficient.

Ownership, governance and financing gates

If ownership is later approved, the consortium owns facility and internal infrastructure, and owns or finances its GPU share; contract construction, specialist maintenance, fuel and diverse external circuits. Development equity requires a bounded study; construction debt requires permits, utility terms, bids and commitments; equipment financing should match energization. The consortium bears delay and underuse unless contracts explicitly transfer them.

Reserve teaching and small-member quotas, release unused reservations, publish marginal-use prices, and use a member committee for admission, scheduling disputes and appeals. Require capped, agreed exit obligations; avoid shifting uncapped costs to smaller institutions.

Evidence and limits: S-CALC is the application model; S-EUROSTAT is the national price series. Country counts and consumption have different scopes and years; see website Evidence for source URLs and limitations. No construction-ready engineering, professional certification or committed financing is claimed. Human source checks and deployment acceptance remain outstanding.